
REE 302 - Machine Design course

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Name	Contribution
Rehand	• Determine the allowable loads acting on the pulley and the gear based on a complete shaft design. Assume the shaft material and use the dimensions provided in Phase I to satisfy the minimum factor of safety against fatigue failure provided in the table below. • Submit a working drawing of the shaft showing all details (Scaled on A4 size sheet).
Hala	• Design the keys of the pulley and the gear. The key design should satisfy the factor of safety provided in the table below. • Estimate the life of both bearings for the shaft speed provided in the table below.

Abstract

This project focuses on determining the allowable loading conditions of a V-belt shaft under specified operating conditions. In Phase II, the objective is to analyze and design the shaft, considering power transmission from a motor through a V-pulley to a spur gear. The main shaft, supported by deep-groove ball bearings, carries a V-pulley at one end and a gear wheel at the other. Forces on the gear and pulley are interrelated, and factors such as pressure angle and rotational speed need to be considered. The minimum factor of safety against fatigue failure (n_f) is a critical parameter. Phase I provides constraints and dimensions for the bearings, gear, and pulley in the axial x-direction. The project entails determining allowable loads, creating a working drawing of the shaft, designing keys for the pulley and gear, and estimating bearing life.

1. Introduction

In the second phase of this project, the focus shifts to the comprehensive analysis and design of the V-belt shaft, emphasizing allowable loading conditions. The V-belt drive system is integral for power transmission, connecting a motor to a spur gear through a V-pulley. The main shaft, pivotal in this configuration, accommodates a V-pulley and gear, supported by deep-groove ball bearings. The forces acting on the gear and pulley are interrelated, with considerations such as pressure angle and rotational speed playing a crucial role. With Phase I providing constraints and dimensions, Phase II tasks include determining allowable loads, creating a detailed working drawing of the shaft, designing keys for the pulley and gear, and estimating bearing life. The minimum factor of safety against fatigue failure, n_f , remains a paramount consideration throughout this design process.

2. Allowable Load based on a complete shaft design

2.1. Material Selection :

The chosen material is **(AISI G1050 STEEL–HR processing)** With a $S_y = 340 \text{ MPa}$, $S_{ut} = 620 \text{ MPa}$ we will design on the worst case scenario with the maximum tensile strength **Brinell Hardness = 179**

Since the tensile strength is less than **(1400 MPa)**, the endurance strength

$$S_e' = 0.5 * \text{tensile strength} = 0.5 * 620 = 310 \text{ MPa}$$

2.2. Calculations:

2.2.1. Force Analysis on Fig1

$$\text{Givens: } \phi = 20^\circ, w = 1300 \text{ rpm}, \eta_f = 2$$

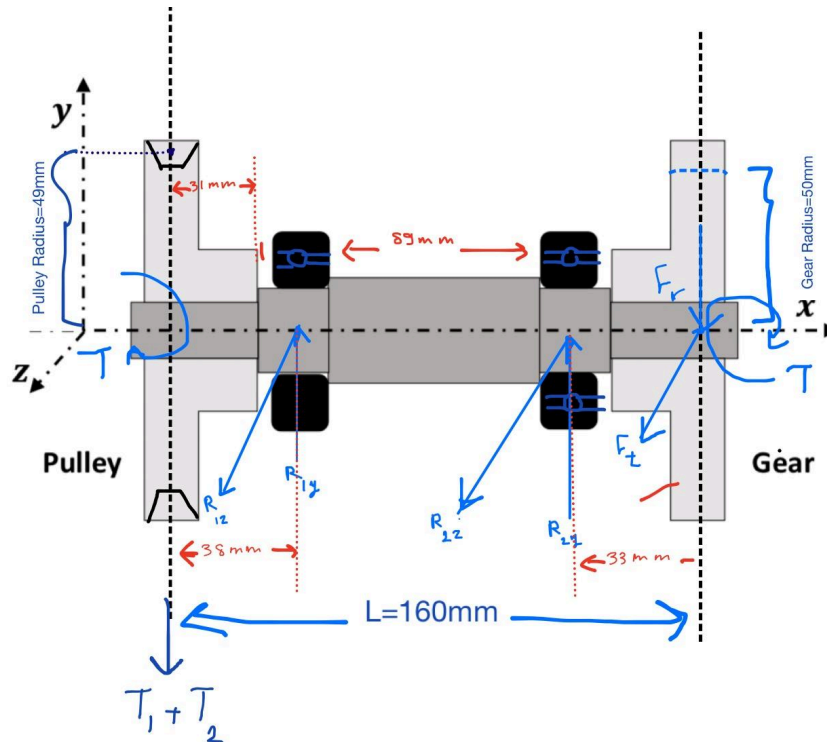


Fig1.Free Body Diagram of the shaft

On the pulley

We start by using

$$T_2 = 0.15T_1 \quad (1)$$

$$T = (T_1 - T_2)R_p \quad \text{as } R_p = 49 \text{ mm} = 0.049 \text{ m}$$

$$T = 0.49(T_1 - 0.15T_1)$$

$$T = 41.65T_1 \quad (2)$$

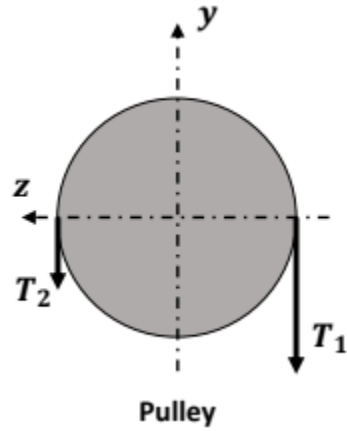


Fig2. Pulley Free body diagram

Acting on the gear

$$T = F_t R_G$$

$$F_r = F_t \tan \phi$$

$$(T_1 - T_2) R_p = F_t R_G$$

$$F_t = \frac{(T_1 - T_2) R_p}{R_G}$$

$$F_t = 0.833 T_1 \quad (3)$$

$$F_r = \frac{(T_1 - T_2) R_p}{R_G} \tan \phi$$

$$F_r = 0.3032 T_1 \quad (4)$$

2.2.2. Reactions in x-y, x-z

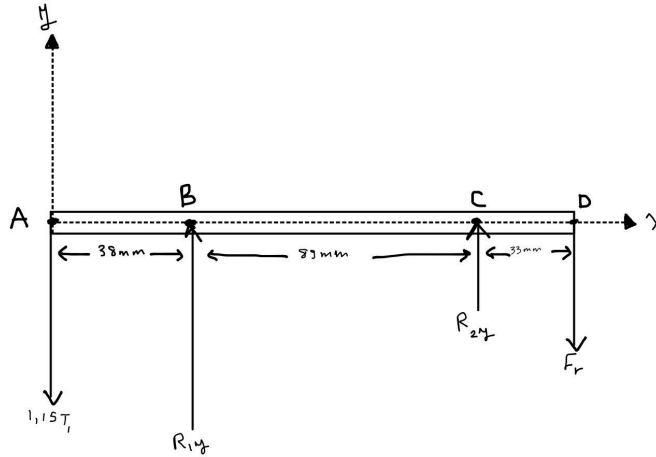


Fig3. FBD on x-y direction

$$T_2 + T_1 = 1.15T_1$$

$$\Sigma F_y = 0$$

$$- 1.15T_1 + R_{1y} + R_{2y} - F_r = 0$$

$$R_{1y} + R_{2y} = 1.45318205T_1$$

$$\Sigma M_D = 0$$

$$122R_{1y} + 33R_{2y} = 184T_1$$

$$R_{1y} = 1.52859T_1, R_{2y} = -0.07541T_1 \quad (5)$$



Fig4. Reactions at Supports(x-y)

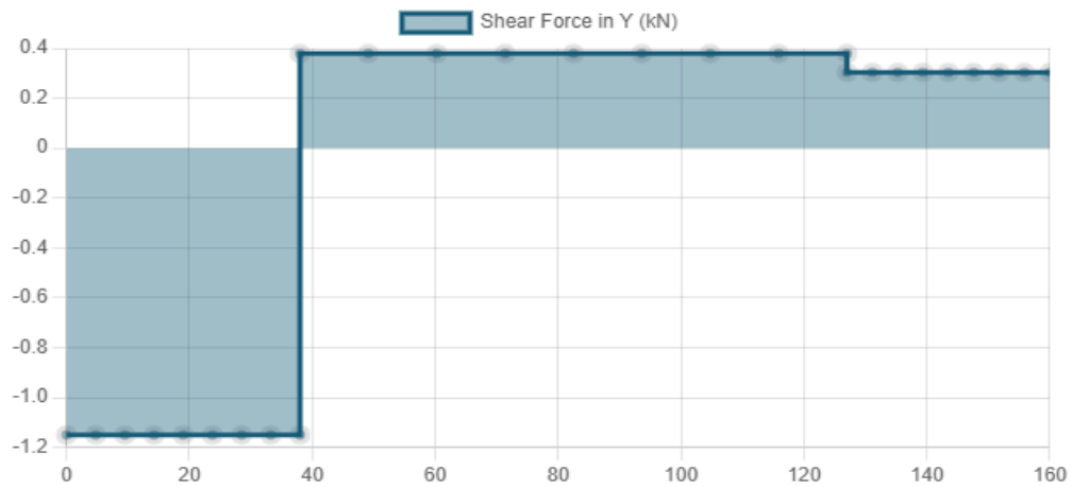


Fig5.Shear Force Diagram(x-y)

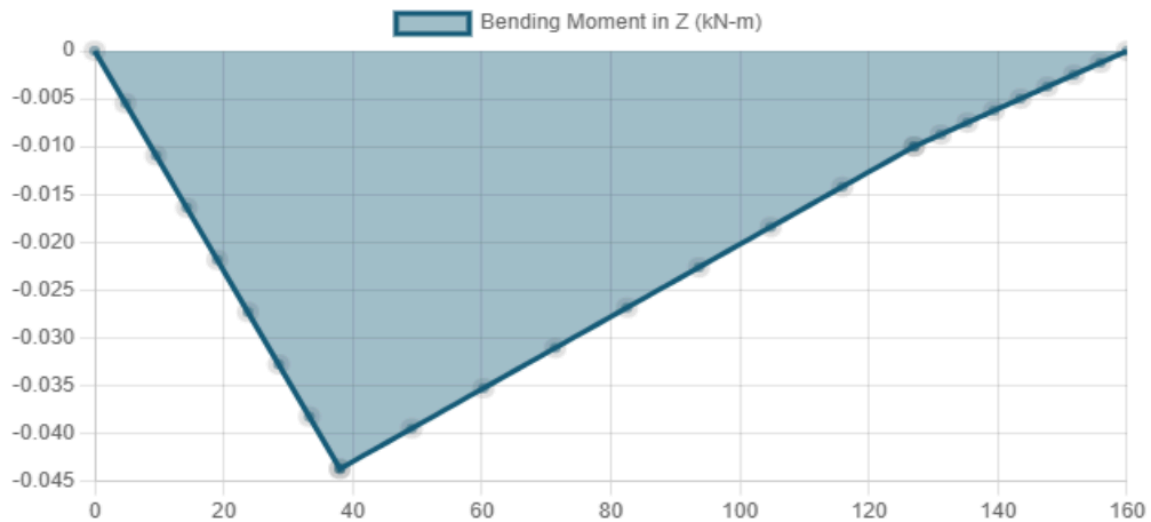


Fig6.Bending Moment Diagram(x-y)

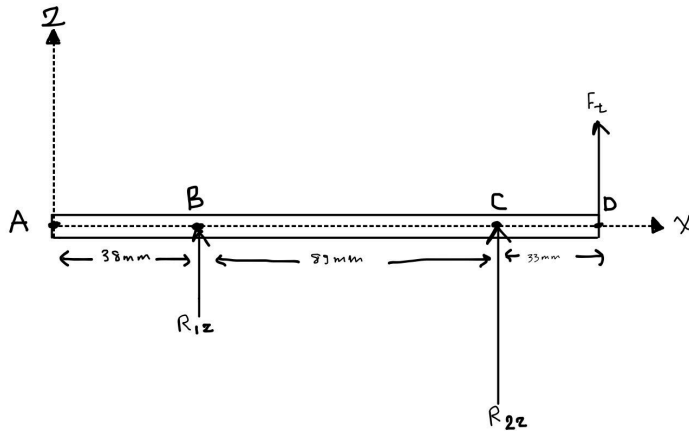


Fig7. FBD on x-z direction

$$\Sigma F_z = 0$$

$$R_{1z} + R_{2z} + F_t = 0$$

$$F_t = \frac{(T_1 - T_2)R_p}{R_g}$$

$$F_t = 0.833T_1$$

$$R_{1z} + R_{2z} = -0.833T_1$$

$$\Sigma M_D = 0$$

$$122R_{1z} + 33R_{2z} = 0$$

$$R_{1z} = 0.30887 T_1, \quad R_{2z} = -1.14865 T_1 \quad (6)$$



Fig8. Reactions at Supports(x-z)

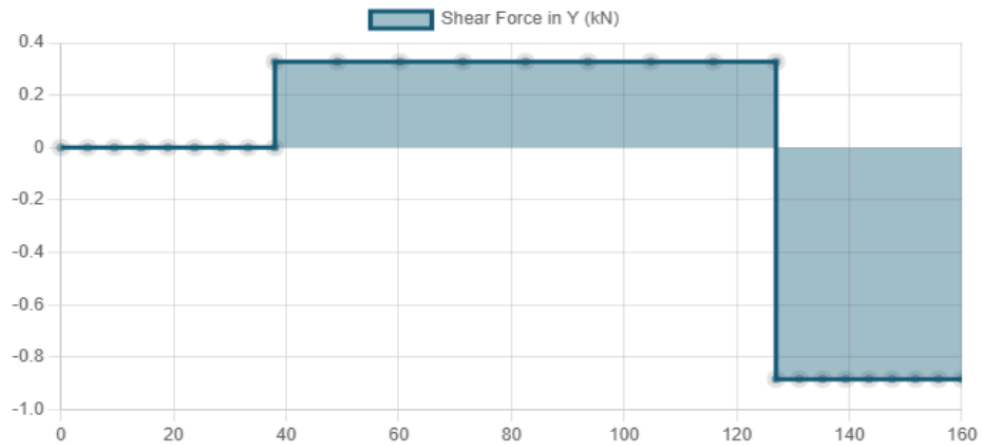


Fig9. Shear Force Diagram(x-z)

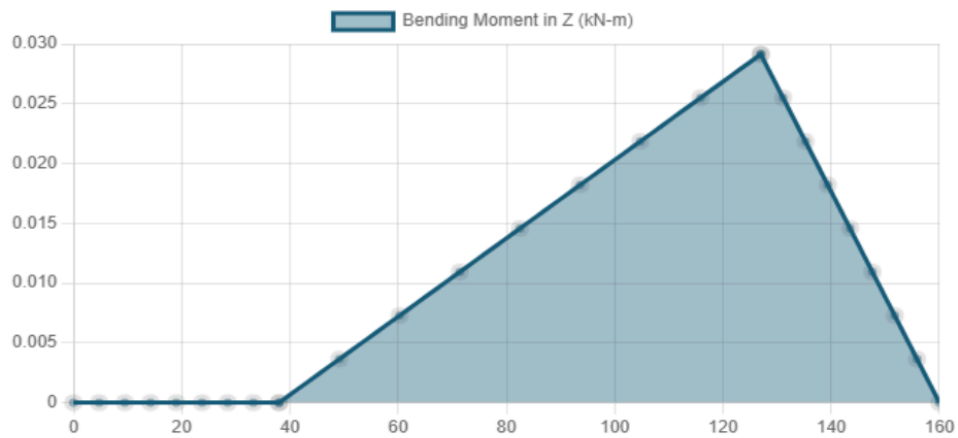


Fig10. Bending Moment Diagram(x-z)

2.2.3. Selection of Maximum Momentum of 4 shoulders

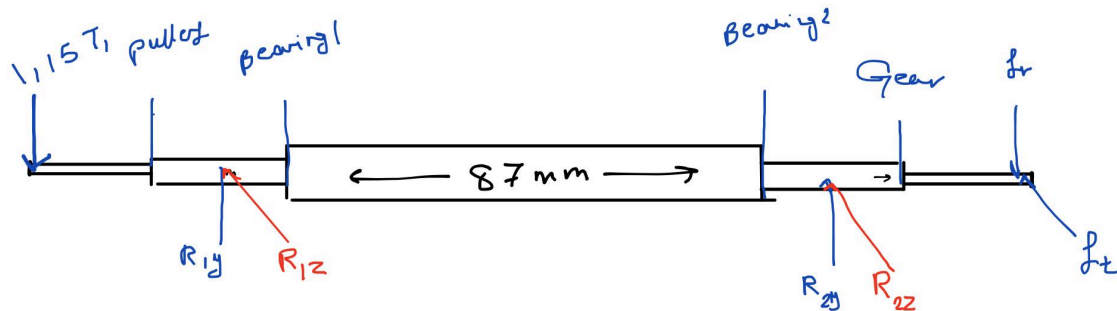


Fig11.FBD on Shaft's Shoulders

Pulley Shoulder

$$M_y = 43.43T_1, M_z = 8.14T_1 \quad M = 44.19T_1 \text{ (max)}$$

Bearing 1 Shoulder

$$M_y = 8.0357T_1, M_z = 7.4769T_1 \quad M = 10.98T_1$$

Bearing 2 shoulder

$$M_y = 23.98T_1, M_z = -19.82T_1 \quad M = 31.11T_1$$

Gear Shoulder

$$M_y = 17.83T_1, M_z = 37.74T_1 \quad M = 41.74T_1$$

2.2.4. To get the value of the unknown(T1)

$$M_a = M_{max} = 44.19T_1 \text{ at a pulley}$$

$$M_m = T_a = 0$$

$$T_m = 41.65T_1$$

$$S_e' = 310 \text{ MPa}$$

$$S_{ut} = 620 \text{ MPa}$$

$$S_y = 340 \text{ MPa}$$

Machined or Cold Drawn:

Surface Finish	Factor a		Exponent b
	S_u , kpsi	S_u , MPa	
Ground	1.21	1.38	-0.067
Machined or cold-drawn	2.00	3.04	-0.217
Hot-rolled	11.0	38.6	-0.650
As-forged	12.7	54.9	-0.758

Fig.12

$$a = 3.04 \quad b = -0.217$$

$$K_a = a(S_{ut})^b = 3.04(620)^{-0.217} = 0.7532$$

$$K_b = 0.9192$$

$$K_c = K_d = K_e = 1$$

$$S_e = K_a K_b K_c K_d K_e S'_e = 214.63 \text{ MPa}$$

$$r = 1$$

$$D/d = 1.11$$

$$r/d = 0.055$$

$$q = 0.83$$

$$q_s = 0.86$$

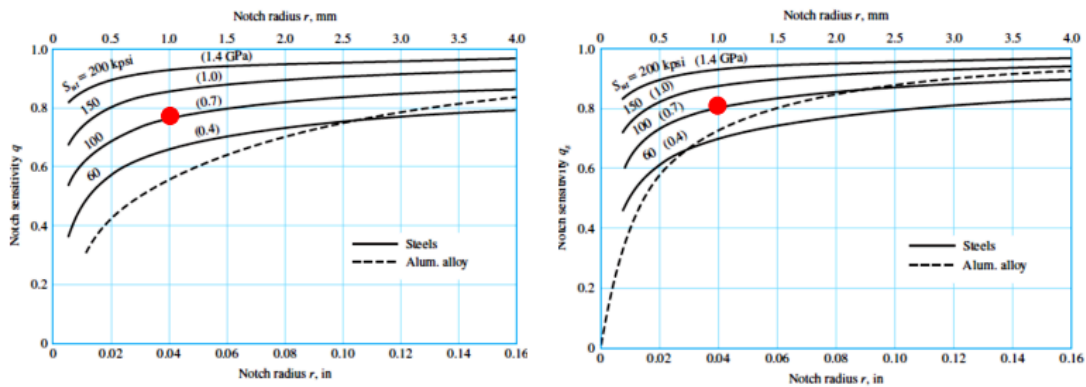


Fig13

$$K_{ts} = 1.25$$

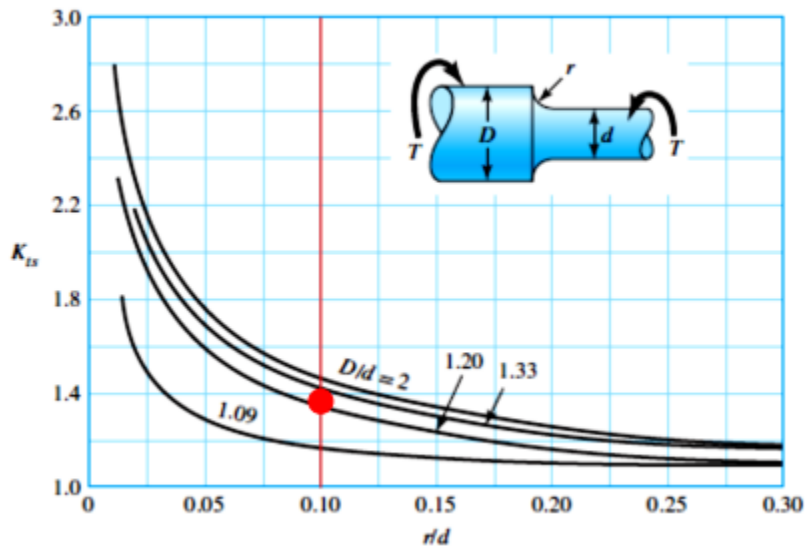


Fig14

$$K_t = 1.75$$

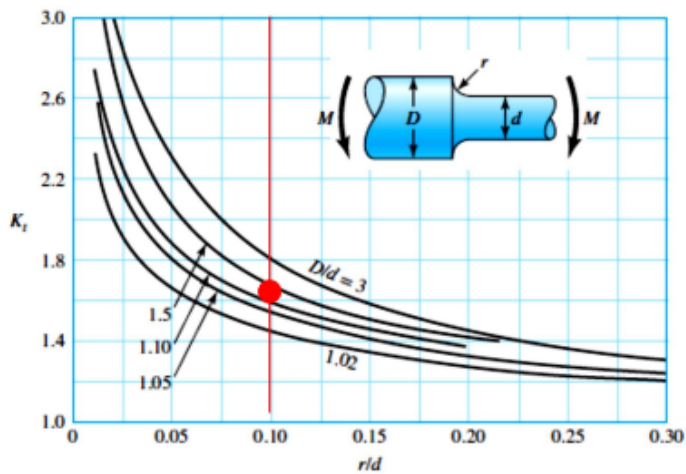


Fig15

$$K_f = 1 + q(K_t - 1) = 1.6225$$

$$K_{fs} = 1 + q_s(K_{ts} - 1) = 1.215$$

$$A = \sqrt{4(K_f M_a)^2 + 3(K_{fs} T_a)^2}$$

$$B = \sqrt{4(K_f M_m)^2 + 3(K_{fs} T_m)^2}$$

$$A=143.397T_1^2$$

$$B=87.649T_1^2$$

$$n = \frac{\pi d^3}{16} \left(\frac{A}{S_e} + \frac{B}{S_{ut}} \right)^{-1}$$

$$T_1=1297.75N$$

$$T_2=194.66N$$

$$F_r = 393.48N$$

$$F_t = 1081.03N$$

$$\sigma_a = K_f \frac{32M_a}{\pi d^3} \quad \sigma_m = K_f \frac{32M_m}{\pi d^3}$$

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1}$$

Shoulder	nf
Pulley	3.883050691
Bearing 1	11.10845432
Bearing 2	3.920630935
Gear	4.110972928

3. Design the keys of the pulley and the gear.

We considered the key material to have a lower strength compared to the main shaft. The chosen key material is **G10200 CD**, characterized by an ultimate tensile strength **Sut= 470 MPa** and a yield strength **Sy= 390 MPa.**”All determined from Fig.12”

$$S_{sy} = 184.64MPa$$

$$T_m = (T_1 - T_2) * R_p = 54.05N.m$$

Since **d=18mm** we can choose **6 mm** square key

As transmitted torque =5.4051Nm

$$F = \frac{T}{r} = \frac{54.051}{0.009} = 6005.67N \text{ where } r \text{ is a radius of the shaft}$$

$$\tau = \frac{S_{sy}}{n_f} = \frac{F}{tL} = \frac{184.64 \cdot 10^6}{2} = \frac{6005.67}{0.006L}$$

$$L = 0.01084m = 10.842mm$$

4. Estimate the life of both bearings for the shaft

$$F_r = 393.48N \quad F_t = 1081.03N, \text{ From Fig18 } C_{10} = 13.5KN, L_r = 10^6 rev$$

$$n_d = 1300rpm \quad F_{D1} = R_1 = 2384.55N, \quad F_{D2} = R_2 = 1588.47N, C_{10} = 13.5KN$$

$$C_{10} = F_R = F_D \left(\frac{L_D}{L_R} \right)^{1/a} = F_D \left(\frac{\mathcal{L}_D n_D 60}{\mathcal{L}_R n_R 60} \right)^{1/a}$$

$$\mathcal{L}_{D1} = \frac{60 \cdot L_R \cdot n_R}{60 \cdot n_D} * \left(\frac{C_{10}}{F_D} \right)^a = 139584.97min = 2326.42 h$$

$$L_{D1} = 60 * \mathcal{L}_D * n_D = 3024346 rev$$

$$\mathcal{L}_{D2} = 472194.48min = 7869.90 h$$

$$L_{D1} = 10230870 rev$$

5. Conclusion

In conclusion, Phase II of the project successfully addressed the critical aspects of determining allowable loading conditions for the V-belt shaft in specified operating conditions. The comprehensive shaft design considered factors such as power transmission, pressure angle, rotational speed, and the minimum factor of safety against fatigue failure. The working drawing of the shaft, presented on an A4 size sheet, encapsulates the meticulous design process. Furthermore, the key design for both the pulley and gear was established to meet the prescribed factor of safety. The estimation of bearing life adds a layer of reliability to the overall design. This phase not only meets the project requirements but also lays the foundation for a robust and efficient V-belt shaft system in practical applications.

6. Appendix

6.1. Working drawing of the shaft

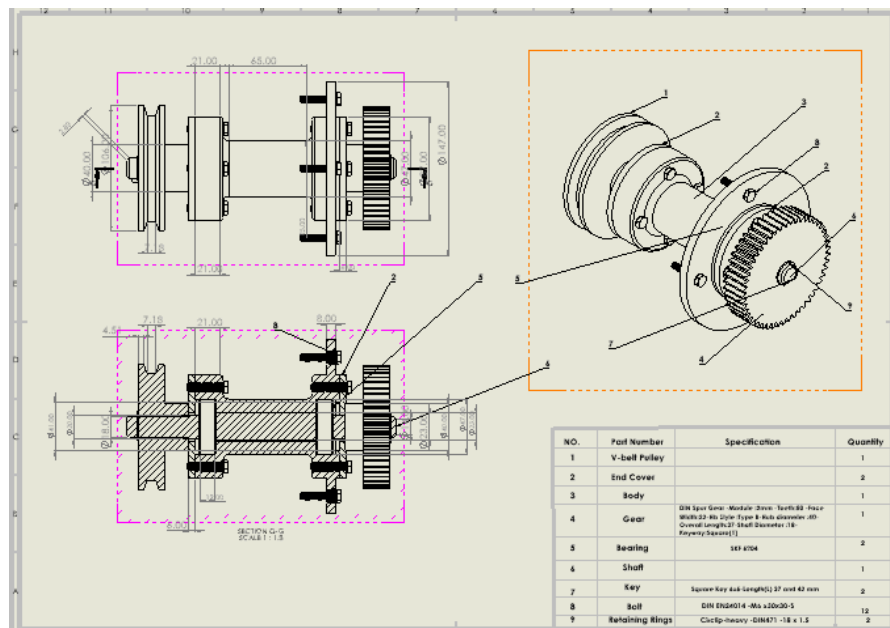


Fig16

6.2. Fig17.Key material (Table A-20)

Table A-20 Deterministic ASTM Minimum Tensile and Yield Strengths for Some Hot-Rolled (HR) and Cold-Drawn (CD) Steels

[The strengths listed are estimated ASTM minimum values in the size range 18 to 32 mm ($\frac{3}{4}$ to $1\frac{1}{4}$ in). These strengths are suitable for use with the design factor defined in Sec. 1-10, provided the materials conform to ASTM A6 or A568 requirements or are required in the purchase specifications. Remember that a numbering system is not a specification.]

1 UNS No.	2 SAE and/or AISI No.	3 Process- ing	4 Tensile Strength, MPa (kpsi)	5 Yield Strength, MPa (kpsi)	6 Elongation in 2 in, %	7 Reduction in Area, %	8 Brinell Hardness
G10060	1006	HR	300 (43)	170 (24)	30	55	86
		CD	330 (48)	280 (41)	20	45	95
G10100	1010	HR	320 (47)	180 (26)	28	50	95
		CD	370 (53)	300 (44)	20	40	105
G10150	1015	HR	340 (50)	190 (27.5)	28	50	101
		CD	390 (56)	320 (47)	18	40	111
G10180	1018	HR	400 (58)	220 (32)	25	50	116
		CD	440 (64)	370 (54)	15	40	126
G10200	1020	HR	380 (55)	210 (30)	25	50	111
		CD	470 (68)	390 (57)	15	40	131
G10300	1030	HR	470 (68)	260 (37.5)	20	42	137
		CD	520 (76)	440 (64)	12	35	149
G10350	1035	HR	500 (72)	270 (39.5)	18	40	143
		CD	550 (80)	460 (67)	12	35	163
G10400	1040	HR	520 (76)	290 (42)	18	40	149
		CD	590 (85)	490 (71)	12	35	170
G10450	1045	HR	570 (82)	310 (45)	16	40	163
		CD	630 (91)	530 (77)	12	35	179
G10500	1050	HR	620 (90)	340 (49.5)	15	35	179
		CD	690 (100)	580 (84)	10	30	197
G10600	1060	HR	680 (98)	370 (54)	12	30	201
G10800	1080	HR	770 (112)	420 (61.5)	10	25	229
G10950	1095	HR	830 (120)	460 (66)	10	25	248

Source: Data from 1986 SAE Handbook, p. 2.15.

6.3. Skf Bearings Fig18

Dimensions

Bore diameter	20 mm
Outside diameter	47 mm
Width	14 mm

Performance

Basic dynamic load rating	13.5 kN
Basic static load rating	6.55 kN
Reference speed	32 000 r/min
Limiting speed	20 000 r/min
SKF performance class	SKF Explorer

6.4. Appendix

7. References

- Shigley's Mechanical Design 11th edition textbook. (n.d.).
- <https://www.skf.com/mena/products/rolling-bearings/ball-bearings/deep-groove-ball-bearings/productid-6204>
- <https://platform.skyciv.com/beam>